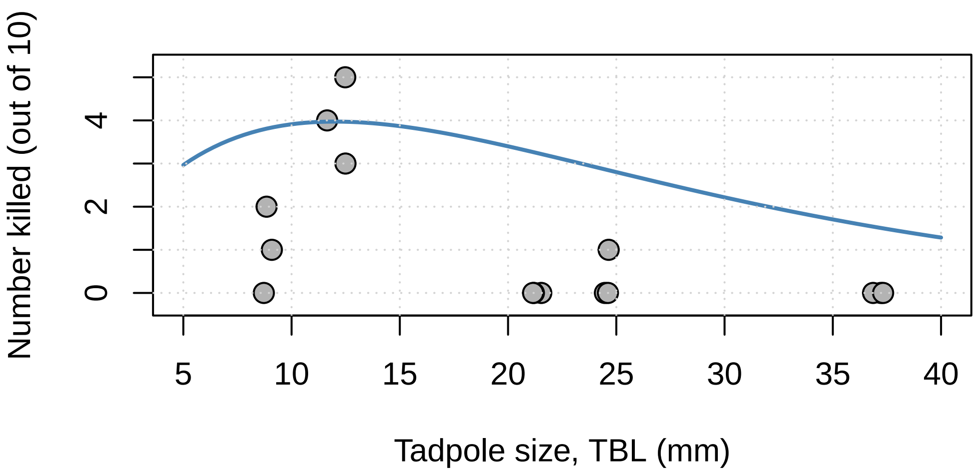
Lab 1, Act 3

Does the evidence actually support our candidate?

NRES 746 – name: ________________________ group: __________________________________

Picking up where we left off

In Act 1, your group diagnosed the tadpole data as hump-shaped and picked the Ricker function as your top candidate. In Act 2, you put specific numbers on it ($a = 0.9$, $b = 1/12$) and derived its peak, $x^* = 12$ mm. Today we put that candidate on trial: does it actually predict the real data well, or does it just *look* plausible?



Part 1: Does the shape earn its keep? (individual, ~10-15 min)

A model earns its keep by predicting better than a "dumb" baseline. One simple way to score any model: for each data point, compute the **residual** (observed − predicted), square it (so overshoots and undershoots both count against the model, and big misses count extra and add up the squares. Smaller total = better fit. This total is often called the **sum of squared residuals (SSR)**.

Model 1: our Ricker candidate. Using $a = 0.9$, $b = 1/12$ from Act 2, fill in the residuals, squared residuals, and total:

TBL (mm)	Observed mean	Ricker predicted	Residual	Squared residual
9	1.00	3.83		
12	4.00	3.97		
21	0.00	3.28		
25	0.33	2.80		
37	0.00	1.53		
			SSR:	

Model 2: the "null" model. The simplest possible model: ignore tadpole size entirely, and always predict the overall average. First, find that average (using either Act 1's data table or the mean of the five group means):

$\bar{y} =$ ____________

Now fill in the rest of the table yourself:

TBL (mm)	Observed mean	Null predicted	Residual	Squared residual
9	1.00			
12	4.00			
21	0.00			
25	0.33			
37	0.00			
			SSR:	

Compare. Which model has the smaller (better) SSR – the Ricker curve, or the null model that ignores body size entirely? Is this what you expected, given Acts 1 and 2?

What does this latest result mean for the tadpole predation model (predation as a function of size)? What do you take away from this result?

Part 2: Find a better fit (group, ~20-25 min)

The Act 2 candidate ($a = 0.9$, $b = 1/12$) was chosen to give a clean critical point, not to fit well! As a group, see if you can do better: try different values of a and b in the Ricker function and drive the SSR down below both models from Part 1.

$$y = a\,x\,e^{-bx}$$

For each candidate (a, b), compute predicted y at all five TBL values (9, 12, 21, 25, 37), then the residuals, squared residuals, and SSR – same procedure as Part 1. Use the space below for your group's scratch work:

Record your group's attempts here (add more rows if you need them):

Attempt	a	b	**SSR**
1			
2			
3			
4			

Your group's best result:

$a =$ ____________ $b =$ ____________ SSR = ____________

We'll compare best-SSR results across groups at the end of class – lowest SSR wins.

Group discussion and revision (~5 min)

Compare your results with the rest of the class – whose best-fit (a, b) won, and how much did that improve on the Act 2 candidate? Feel free to revise any of your answers above based on the discussion (it's fine to cross out and rewrite – I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion ("muddiest points") or topics you want to review or dust off going forward.